CasePath: Agents That Read, Rules That Decide

Requesting Evidence for Insurance Claims with Executable Workflows

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Case handlers must request evidence before a file settles the deciding facts. In CasePath, a model reads each such fact three times and quotes the file. Code admits a value only when the readings agree, executes the workflow's rules, and holds every request that depends on an open fact. We tested a frozen commit on sealed synthetic claims under a pre-registered plan. On tenancy it made 0.31 strict unsafe errors per claim, against 2.14 for a single prompt with the same rules and model and 1.86 on a frontier model; on theft, 0.58 errors per pair against 1.74. These three hypotheses were confirmed; its theft lead over self-consistency (1.21) missed significance, so a fourth failed and a fifth went untested. After the test we told a single prompt the same stop and hold rules in plain words. It matched CasePath on tenancy (0.35 on the same model, 0.19 on the frontier model) and made no error on the frontier model where a claim denies its defining event, where CasePath held needlessly. CasePath kept its lead where a fact is tentatively worded (0.52 against 1.25 deviations per claim), where a supplied document is insufficient, and on theft (0.58 against 1.68). The registered tenancy result therefore measures mostly what the prompt was told. What the design adds is fewer wrong holds, checks of supplied documents, and requests that stay stable when one sentence changes. Code, synthetic data and analysis will be released.

CCS Concepts: • **Applied computing** → **Business process management**; *Law*; • **Computing methodologies** → **Knowledge representation and reasoning**; *Natural language processing*.

Additional Key Words and Phrases: LLM agents, evidence requests, claims handling, rule execution, three-valued logic, supervaluation, neuro-symbolic systems, pre-registered evaluation, design science

1 INTRODUCTION

A single sentence in a claim can decide which documents a case handler should request. A tenant reports mould, and the insurer's rules require a medical confirmation if the mould has affected someone's health. One version of the claim says "the paediatrician suspects a link with the mould"; another says "we do not know whether it has anything to do with the mould". For the first, the handler should request the confirmation. For the second, the handler should hold it and ask one question (Figure 1). A request made too early costs the claimant and the handler time, a document dropped too early weakens the decision, and an insurer must be able to show why it asked.

This is one instance of a class of problems: deciding which evidence to request while the file leaves deciding facts open. We treat it as design-science research [17, 19]: we build an artefact for the class, state design principles with their boundary conditions, and evaluate the artefact.

The idea is a division of labour. A language model reads each deciding fact three times and must quote the file for every value. Code admits a value only when the readings agree and the quotation is found, executes the rules of a workflow, and holds every request whose need depends on a fact the file leaves open. A single prompt that applies the rules records no fact values that code could check, and does not keep a fact the file omits apart from one it denies or leaves open.

We implemented the idea in CasePath, a product that prepares legal-protection claims for a case handler, and tested it at a frozen commit on sealed synthetic claims under a pre-registered plan (Section 6). We count a *strict unsafe error* when a design requests a document the rules do not

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The claim states the health effect

claim reader · source view

Our daughter has been coughing for weeks; the paediatrician suspects a link with the mould. I do not have a medical report yet.

The claim leaves the health effect open

claim reader · source view

Our daughter has been coughing for a few weeks; we do not know whether it has anything to do with the mould and have not seen a doctor.

Document	Health effect stated	Health effect open
Lease contract	present	present
Defect notice	request	request
Proof of notification	request	request
Dated photographs	request (file unusable)	request
Landlord response history	request	request
Humidity and temperature log	request	request
Medical confirmation	request	hold; question: does the claim allege effects on anyone's health?
Caretaker correspondence	optional, not requested	optional, not requested

Fig. 1. The running example: one changed sentence turns the request for the medical confirmation into a hold with one question; the other requests are unchanged. Top: the source view with the quoted words highlighted (unedited captures of the product at commit 6a72159c on two synthetic claims with the tenancy defect workflow; cropped). Bottom: the checklist that code evaluated for each claim, typeset from the stored results of the same two runs (`paper/figures/ui/s4/result-stated.json` and `result-open.json`). The lease is attached to both claims.

need, requests one whose need still depends on an open fact, leaves out or hands over a needed document, or marks as not needed a document whose need depends on an open fact. On tenancy claims, CasePath made 0.31 strict unsafe errors per claim, against 2.14 for a single prompt with the same rules on the same, cost-efficient model and 1.86 on a frontier model (registered; $p = 0.0002$ and $p = 0.0008$). It held 97% of the documents the reference holds, against 48% and 35% (registered). Added after the test, a prompt told the same stop and hold rules in plain words matched it overall, with 0.35 strict unsafe errors per claim on the same model and 0.19 on the frontier model; the lead that remains is on open branch facts, insufficient documents and theft (Section 6.6). On theft pairs with an unresolved fact it made no premature request and changed nothing else in 94% of them (registered).

On the cost-efficient model, CasePath had a lower mean error than every single-prompt design and every strategy from prior work in both data sets (registered). The registered tests confirmed this against the single prompt in both data sets and against two of three prior-work strategies on theft. Against self-consistency, a majority vote over several samples, the theft lead (0.58 against 1.21 errors per pair) gave $p = 0.0625$, and that hypothesis failed. On the frontier model the product ran once, in its checklist-only setting, without a hypothesis. It made fewer tenancy errors and had no advantage on theft: 0.83 errors per pair against 0.95 for the frontier prompt and 0.67 for that prompt with the rules before review (registered secondary). It also made more theft errors than the product on the cost-efficient model (0.83 against 0.58, $p = 0.0781$, registered secondary). An error decomposition added after the test puts the difference in the frontier model's readings and holds, and its stored readings trace it to two broadly worded rule conditions. Section 7.1 states the scope of these results.

The article makes three contributions:

(1) A decision rule that admits only agreeing, quoted readings of the deciding facts and holds exactly the documents whose need depends on an open fact, with its properties and their scope (Section 3).
(2) CasePath, the artefact, and the design knowledge from building and testing it (Sections 4, 5 and 7).
(3) A pre-registered evaluation on sealed data against single-prompt and prior-work strategies, with checks added after the test: an error decomposition, a budget-matched comparison, a clean-room re-derivation of the references and their sensitivity to the reading of the rules (Section 6).

The code, the research workflows, the synthetic data and the analysis will be released [32].

2 BACKGROUND AND RELATED WORK

CasePath combines rules executed by code, models that read text, and an explicit value for what is not known. Table 1 compares the closest systems.

Design science. Design-science research builds an artefact for a class of problems, evaluates it [19, 42], and states design principles as its knowledge contribution [17]. Action design research builds and evaluates inside the organisation [47]; this article evaluates on synthetic claims.

Rules over text. Law and administration have executed rules since the British Nationality Act was written as a logic program [48]; Catala [36], the Decision Model and Notation standard (DMN) [39] and rules as code in government [37] continue the line. They take typed facts and treat a fact not supplied as false [44] or send the decision to a default branch; CasePath reads facts from text and keeps them unresolved unless a stated presumption applies. Neuro-symbolic systems have a model translate each problem into logic that a solver executes [13, 40, 41, 57], and legal systems compute an answer, such as a tax liability, from a complete representation of a case [20, 24, 27]; CasePath executes one reviewed workflow and must act before the case is complete. The closest system, LDPC, available as a preprint [9], answers yes/no questions as true, false or maybe and evaluates a model-written formula in strong Kleene logic, asking about a maybe; Proposition 3.2 shows what that evaluation costs when a workflow's branches meet again.

Case management and process models. Case management [38], declarative process models [43] and artifact-centric process models [22] let a case's data decide which tasks and documents apply. They assume the data are entered; CasePath reads them from the file and acts while some are unknown. Claims handling is such a knowledge-intensive process [5, 7, 8], and process research asks how language models can act within explicit constraints [6, 26, 30, 33, 51].

Agents that apply the rules. Agents usually follow a policy placed in their prompt, as τ-bench, FlowBench and procedure-following benchmarks evaluate [35, 54, 55]; Chain of Logic has the model evaluate a rule's elements [49], ReAct interleaves reasoning with tool calls [56], and self-consistency takes the majority of several runs [52]. In these designs the model applies the rules, and repeated runs differ [1, 2]. Section 6 tests CasePath against a single prompt with the rules, self-consistency, a prompt in the style of Chain of Logic that states the rules' semantics (Model executes rules), and ReAct; it does not run Chain of Logic itself or the benchmarks' agents.

Reasoning over partial information. Three-valued model checking gives an answer that holds for every completion of a partial state space [3], partial models in software engineering reason over all concretisations of an uncertain model [11], and supervaluation [50] and certain answers over incomplete databases [23] do the same for sentences and queries. Exact deferral applies this idea to

Table 1. Prior systems and CasePath. Yes, partly or no as stated in the cited work; –: not addressed there. Columns: (1) rules executed by code; (2) explicit unresolved value; (3) quotation required; (4) deferral exact. [a]The model writes the logic for each item. [b]For the facts that decide the checklist or the routing, within the scope stated in Section 3.

System	1	2	3	4
Rules as code, DMN, Catala [36, 37, 39, 48]	yes	no	no	no
Policy in the prompt [35, 54, 55]	no	–	–	–
Chain of Logic [49]	no	–	–	–
Logic-LM, LINC, SatLM, PAL [13, 40, 41, 57]	yes[a]	no	–	no
Statutes as programs, extracted facts [20, 24, 27]	yes	no	no	no
LDPC (preprint) [9]	yes[a]	yes	no	no
Case management, declarative and artifact-centric processes [22, 38, 43]	yes	–	–	–
Asking and abstaining [25, 34, 58]	no	partly	–	no
Evidence-obligation ledger (preprint) [28]	partly	yes	partly	–
CasePath (this article)	yes	yes	yes[b]	yes[b]

the documents a workflow needs; what is new is the application and the reconvergence case of Proposition 3.2, not the semantics.

Presumptions. CasePath's presumptions are defaults in the sense of default logic [45], fixed per fact in the workflow with a source clause; unlike the Carneades model [16], the product does not reason about the burden of proof.

Asking, agreement and quotation. Selective prediction abstains when a model's confidence is low [4, 15, 25, 53], and interactive agents ask for missing information before they act [34, 58]. CasePath asks only when an open fact can change a document, a deterministic special case of the value of information [21]. Semantic entropy measures uncertainty from the agreement of sampled answers [12, 31], and citation benchmarks score whether cited sources support a model's statements [14]; CasePath makes agreement and a found quotation hard conditions, checked by code. The nearest agent design, a preprint that is not peer reviewed, keeps a ledger of evidence obligations in code, tied to their sources [28].

Legal benchmarks and supervision. Our test measures how a design applies fixed rules to synthetic files, not legal reasoning as LegalBench does [18]. The EU AI Act's high-risk class covers life and health insurance, not evidence requests in legal-protection claims [10].

3 THE DECISION RULE

The decision rule divides the work: a model reads the file and gives each fact a value, and code applies the workflow to those values. This section defines the rule and states four results with their scope; the proofs are in Appendix A.

We carry the mould claim of Section 1 throughout. One file says: "Our daughter has been coughing for weeks; the paediatrician suspects a link with the mould. I do not have a medical report yet." Another says: "…we do not know whether it has anything to do with the mould and have not seen a doctor."

Table 2. Admission of a fact that decides the checklist or the routing, at commit 6a72159c. "Value" means a true or false answer, or a value, with its quotation found. A fact left unresolved holds the documents whose need it can change.

What the readings give	No presumption	Presumption of value v
Every reading gives the same value x	x is admitted	x is admitted
Readings give different values	unresolved; marked for a person	unresolved; marked for a person
Some readings give x, the others no value	unresolved	v if $x = v$; otherwise unresolved
No reading gives a value; one or more answer open with words found	unresolved	unresolved
No reading gives a value; none answers open with words found	unresolved	v

3.1 Workflows and facts

A workflow states an organisation's rules for one kind of claim. It names a finite set C of facts, the conditions its rules depend on, such as *health*: the mould has affected someone's health. A fact is true (T), false (F) or unresolved (U).

The workflow arranges steps in an acyclic graph $G = (N, E)$ with one entry step. Each step n has a condition c_n, and each transition $(u, n) \in E$ has a condition c_{un}. Each document type d is needed at a set of steps $L(d) \subseteq N$ under its own condition c_d. Conditions are formulas over facts built with *and*, *or* and *not*. We call such a workflow a *flow workflow*: its transitions only enable or branch to a step, with no relation that blocks, suppresses or waits for another step.

The output, a checklist, marks each document type request, hold, not needed or present, and lists as questions the facts on which held documents wait. Whether a document is on file is decided by two model judgments (Section 4.1); the propositions take it as given.

The product also has facts that take one of a list of values, a date or a text; for exact deferral such an open fact takes each listed value, or each value the conditions name and one for any other. In the tested workflows only the fact that selects the workflow is not two-valued. Completions treat facts as independent, so for related facts exact deferral can hold a document that every completion consistent with the relation needs.

3.2 Readings, admission and presumptions

A reading is one model pass over the file that answers every fact true or false with a quotation, open (the file says it is not known, quoting the words) or not stated. CasePath makes three readings by default, each a separate request. A quotation is *found* when, after soft hyphens are removed, words split by a hyphen at a line break are joined and white space is collapsed, it occurs letter for letter, with letter case kept, on one page of a file; failing that, a second match also equates the typographic variants of quotation marks, apostrophes, dashes, ligatures and spaces. Nothing else is folded, and a quotation shorter than eight characters is never found. A value whose quotation is not found, even after one request to quote again, counts as no value.

Table 2 gives the admission rule of the tested commit for the facts that decide the checklist or the routing: those in a transition, in a document's condition or in the workflow's selection condition. Code, not the model, applies it. Other facts are recorded from the first reading alone and do not enter the evaluation.

For the first file every reading gives *health* = T and quotes "Our daughter has been coughing for weeks; the paediatrician suspects a link with the mould", which is found, so T is admitted. For the

second every reading answers open, quoting “we do not know whether it has anything to do with the mould”.

A file that never mentions a fact does not express doubt about it. A *presumption* states, with a clause from the organisation’s sources, what an unstated fact is taken to be. The checklist records a presumed value as a presumption. The tenancy workflow presumes that the mould has affected no one’s health when the file never mentions health. In the second file the readings answer open, so *health* stays U.

Admitted values and presumptions form an assignment $v : C \to \{\mathsf{T}, \mathsf{F}, \mathsf{U}\}$. A *completion* of v is an assignment $v' : C \to \{\mathsf{T}, \mathsf{F}\}$ that agrees with v wherever $v(f) \neq \mathsf{U}$.

3.3 Evaluation

Code evaluates conditions in strong Kleene logic [29]. With the order $\mathsf{F} < \mathsf{U} < \mathsf{T}$, *and* takes the minimum, *or* the maximum, and *not* swaps T and F. A step is active when it is the entry step or is reached from an active step through a transition that holds, and its own condition holds:

$$a(n) \;=\; \Big[\,\mathrm{entry}(n) \;\vee \bigvee_{(u,n)\in E} \big(a(u) \wedge v(c_{un})\big)\Big] \;\wedge\; v(c_n). \tag{1}$$

A document is needed when a step that needs it is active and its own condition holds:

$$r(d) \;=\; \Big[\bigvee_{n\in L(d)} a(n)\Big] \;\wedge\; v(c_d). \tag{2}$$

A document with $r(d) = \mathsf{F}$ is not needed; one with $r(d) = \mathsf{U}$ is held; one with $r(d) = \mathsf{T}$ is present if the file contains it and requested otherwise. We write r_v when the assignment matters. Equations (1) and (2) are the semantics of flow workflows. All four tested workflows are flow workflows: under every complete assignment, the product’s own walk at the tested commit lists the documents these equations give (Appendix A says how we checked).

Proposition 3.1 (Soundness relative to admitted facts). *Let the workflow be a flow workflow over two-valued facts. If $r_v(d) \in \{\mathsf{T}, \mathsf{F}\}$, then $r_{v'}(d) = r_v(d)$ for every completion v' of v.*

Whatever the open facts turn out to be, a request stays needed and a document marked not needed stays unneeded. Learning an open fact’s value can turn a held document into a request or into not needed, but never reverses a request or a not needed. Two-valued evaluation lacks this property. Reading U as F, the closed-world assumption [44], drops documents that an open fact may require; reading U as T requests them too early.

Proposition 3.2 (Strong Kleene evaluation over-holds). *Even in a flow workflow over two-valued facts, strong Kleene evaluation can hold a document that every completion of v needs.*

Strong Kleene evaluation forgets that two branches of a fork meet again. The tenancy defect workflow branches on *health*: to “Escalate health risk” if it is true, directly to “Dated defect timeline” otherwise, and the first leads to the second. With *health* = U, Equation (1) makes the timeline step U, and every later document whose own condition is not false is held. Under both completions the step is active (Figure 2).

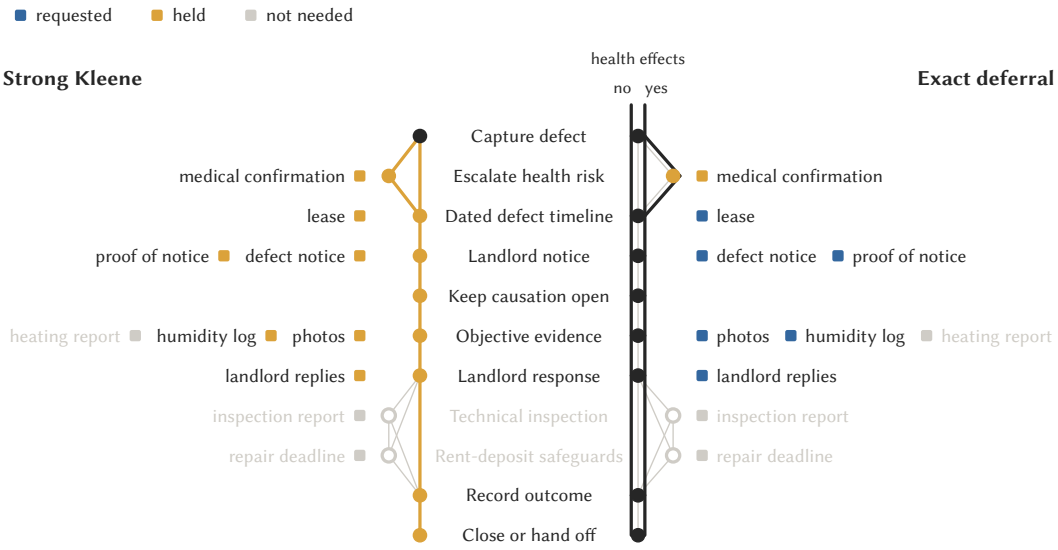


Fig. 2. The tenancy defect workflow with *health* unresolved, mould true, the other facts false and no document on file. Strong Kleene evaluation (left) holds every document after the branch whose own condition is not false; the heating report, the inspection and the repair deadline are not needed because their conditions are false. Exact deferral (right) holds only the medical confirmation, on which the two completions disagree. Because nothing is on file, the lease is requested here; in Figure 1 the claims attach the lease, so the product lists it as present and does not request it.

3.4 Exact deferral

Exact deferral evaluates each document under every completion [50], as certain answers do for an incomplete database [23]:

$$\hat{r}(d) \;=\; \begin{cases} \mathsf{T} & \text{if } r_{\nu'}(d) = \mathsf{T} \text{ for every completion } \nu' \text{ of } \nu, \\ \mathsf{F} & \text{if } r_{\nu'}(d) = \mathsf{F} \text{ for every completion } \nu' \text{ of } \nu, \\ \mathsf{U} & \text{otherwise.} \end{cases} \tag{3}$$

The checklist is formed from $\hat{r}$ as from r. An open fact becomes a question when changing its value changes a held document or a step.

By definition, exact deferral holds a document exactly when two completions of ν disagree on it, and a document it requests or marks not needed has that status under every completion; this needs only that a document's need is defined under each complete assignment.

Proposition 3.3 (Exact deferral and strong Kleene evaluation). *In a flow workflow over two-valued facts, exact deferral never holds a document that strong Kleene evaluation requests or marks not needed.*

Exact deferral holds what the open facts can change and nothing else. In Figure 2 it requests the lease, the defect notice and the later documents, holds the medical confirmation and asks whether the mould has affected anyone's health. It costs one evaluation per completion. Above eight open facts or 256 combinations of answers, the product returns the plain walk instead, which stops at the first transition whose condition is open and holds each document whose own condition is open. In Study 3, tenancy claims had at most 3 open facts and got exact deferral. Exact deferral never ran

on theft: every theft text had 9–28 open facts and got the plain walk, which on the theft workflow gives what exact deferral gives (Appendix A). Study 1's research implementation fell back to strong Kleene evaluation above twelve open facts, which its claims never reached.

3.5 Reading errors

A reading can err in two ways: it can leave a settled fact unresolved or give it the wrong value. Let v^* be the true, complete assignment. An *unsafe error* is a request for a document that v^* does not need, or a document marked not needed that v^* needs.

Proposition 3.4 (Reading errors). *Let the workflow be over two-valued facts, and take whether a document is on file as given. (a) If $v(f) = v^*(f)$ for every fact with $v(f) \neq \mathsf{U}$, exact deferral makes no unsafe error. (b) If $v(f) \neq v^*(f)$ for some fact with $v(f) \neq \mathsf{U}$, every document whose $\hat{r}_v$ is T or F but differs from r_{v^*} is an unsafe error.*

An unresolved value costs holds and a question; a wrong value that passes admission produces an error the evaluation cannot detect. A presumption that the file's facts contradict is such a wrong value. Without a model, we computed exactly the expected outcome over the 150 tenancy claims of Study 1 when each fact is misjudged independently with probability 0.1, once with every misjudgement unresolved and once with every misjudgement the opposite value (Appendix A, Figure 5). Unresolved values became expected holds under exact deferral and no unsafe error; wrong values became expected missed documents and unnecessary requests.

Admission and agreement exist to turn wrong values into unresolved ones. In Study 3, an analysis added after the test found that over 1344 tenancy fact values (facts × claim-runs) 9 were admitted with a wrong value, each time with all readings agreeing. Of these, 6 admitted a denied defining event as having happened, out of 72 such claim-runs, in which the readings differed 12 more times. In the decomposition of Section 6.3, reading errors on the denied-event claims make up about half (49%) of CasePath's strict unsafe tenancy errors (Supplement, Table S35).

4 CASEPATH

The artefact is the core of CasePath at commit `6a72159c` (source-tree hash `03048020. . .`): readings with quotations, admission by agreement, document judgments, and code that executes the workflow with exact deferral. It takes a claim's files and a workflow and returns a checklist that marks each document as requested, held with a question, present or not needed.

4.1 The core

The core runs four steps in a fixed order. A *routing* step answers the routing facts with quotations from the files; code selects the workflow whose condition the admitted values satisfy and hands the claim to a person when none or several apply. Three *readings*, each a separate model request, then answer every condition of the workflow with a value and a quotation, and code admits values as Table 2 prescribes. If an admitted value contradicts the chosen workflow, code routes once more; a second contradiction hands the claim to a person. Two *document judgments* decide whether each needed document is in the files: a first model call names the file and quotes each item the need lists, and a second, independent call must confirm the same file and items, or the document stays requested. Last, code evaluates the workflow by exact deferral over the admitted facts, the presumptions and the confirmed documents (Section 3.4). No model can change a deciding fact after the readings, and no document becomes present without both judgments. For a claim of one file, the tested core (Section 6.1) skips the document judgments and counts the needs as missing.

Listing 1. The tenancy defect workflow (`defect_mold_heating.json`), trimmed with ellipses and with keys reordered for reading; nothing is added.

```
{"id": "defect_mold_heating", "applies_when": {"fact": "dispute_family", "is": "defect_mold_heating"}, ...
 "facts": [{"id": "health_effects", "type": "boolean", "question": "Does the claim allege effects of the defect
        on anyone's health?", ...}, ...],
 "documents": [{"id": "medical_confirmation", "required_when": {"fact": "health_effects", "is": true},
        "must_show": ["health findings", "causal limits"], ...}, ...],
 "steps": [{"id": "dh_intake", "next": [{"to": "dh_safety", "when": {"fact": "health_effects", "is": true},
        ...}, {"to": "dh_scope", "when": {"fact": "health_effects", "is": false}, ...}], ...}, ...],
 "presumptions": [{"fact": "health_effects", "applies": "unstated", "value": false, "source_ref":
        "policy-defect_mold_heating-v1#...clause-silence", ...}, ...], ...}
```

4.2 Workflows

A workflow is a data file in the format `casepath.workflow/1`. Every fact, step, document need and presumption must cite a source, or the file is rejected. Steps form a graph with conditions on transitions. A document need has a condition (`required_when`) and may list what the document must show (`must_show`). A presumption states what an unstated fact is taken to be and cites the clause that says so. Listing 1 shows the tenancy defect workflow of the running example.

5 HOW THE DESIGN WAS FOUND

The design came from one cycle of building and evaluating on synthetic data, with three pre-registered studies.

Two earlier configurations were each tested in a pre-registered study, and neither made fewer errors than Rules in context, a single prompt that applies the rules. The Study 1 configuration made one reading, had a second model call accept or reject each value, and evaluated a theft workflow that a model had extracted from insurance law and policy terms. With the frontier model it was no better than Rules in context: 38 against 22 wrong theft changes ($p = 0.211$) and 0.081 against 0.063 unsafe errors per tenancy claim, by Study 1's measure ($p = 0.250$). Study 2 tested a second configuration on sealed new data and confirmed none of its four primary contrasts. A first redesign, which withheld every rule an automatic check could not repair, was dropped on development data before Study 2. A repair without reference answers, in which a model checked each extracted rule against its cited passages and edited flagged rules, did not lower errors in Study 2, and learning from handler corrections did not separate the configuration from Rules in context. Because code applied the rules, each error could be traced, post hoc, to a rule or a reading (Supplement, Table S13). Supplement, Sections S1 and S2 give both studies in full.

Four changes remain and form the decision rule of Section 3: a reviewed workflow, readings that must agree, presumptions and exact deferral. The theft workflow was reviewed rule by rule. Two AI editor sessions checked each of the 98 extracted theft rules against its cited passages, independently; a change was adopted only when both made it, or one made it and the other approved it (a four-eyes rule). They changed 9 rules and queued 14 for a jurist. The nine theft conditions that change between the two texts of a pair are the same in all three studies. No claim here rests on an unplanned pilot on the ShARC benchmark [46] (Supplement, Section S6).

6 EVALUATION (STUDY 3)

We tested CasePath at a frozen commit against designs that give a model the same files and the same rules, on sealed synthetic data under a plan fixed in advance. We call this test Study 3. It has two data sets, theft and tenancy, each with its own workflow and measures. Section 6.2 gives the registered outcomes. Sections 6.3 to 6.8 add the plan's registered secondary results and analyses

Table 3. Timeline of Study 3, in UTC, from the commit dates of the repository and the run log.

Time	Event	Record
9 Nov, 22:37	`fresh/` sealed (data of Study 2, development data of Study 3)	commit 1c84cb3e
10 Nov, 17:05	`fresh2/` sealed	commit 5b45aa24
10 Nov, 19:17	first draft of the Study 3 plan	commit 71eaa9eb
10 Nov, 19:51	16 probe claims, runs and diagnosis committed	commit 6c7f4c89
10 Nov, 21:26	plan in final draft, five hypotheses	commit 984d2991
10 Nov, 21:51	hypotheses reordered (Section 6.2)	commit 482873e3
10 Nov, 21:53	open rule `any` committed to the product	commit a0d6211d
10 Nov, 22:18	product frozen at the tested commit	commit 6a72159c
10 Nov, 22:29	plan, specification, driver and analysis frozen	commit fd35e438
10 Nov, 22:30	run starts; all preflight checks pass	run log
11 Nov, 04:17	fifth invocation completes the run	run log
11 Nov, 04:18	analysis run once on the stored outputs; results committed	commit 48488138

added after the test. We mark the latter with a dagger (†) in headings, tables and figures; none is registered, none changes a hypothesis's status, and their p values are descriptive.

6.1 Data, designs and freeze

The test data, `fresh2/`, were built and sealed before any design ran on them (Supplement, Section S10). Theft has 8 new stories: 72 *changed-fact pairs*, two texts that differ in one sentence that changes one of nine conditions, and 48 *unresolved-fact pairs*, in which the second text leaves that condition open. Tenancy has 96 new claims in 24 *families*; a family is four claims of one dispute type that share a combination of true facts. Half are in English and half in Swiss Standard German. Each tenancy dispute has a *defining event*, such as receiving a written rent-increase notification. The plan defines three subsets from the reference: 24 *denied-event* claims, which say the event has not happened; 9 *open-event* claims, which leave it open; and 63 *ordinary* claims, of which 23 leave another fact open. The test set was designed to include the denied- and open-event claims.

The tenancy references were produced by the engine CasePath implements, from each claim's designed fact states. All designs except the two original-rules designs received the theft rules as reviewed (Section 5). The open rule (below) was set before the freeze; Appendix B records how. Table 3 gives the dates.

All designs reached the models through OpenRouter's chat-completions API: gpt-6-luna, the *cost-efficient model*, and gpt-6.1-terra, the *frontier model*; no snapshot is recorded. Table 4 lists the designs. *Model executes rules* is the plan's design in the style of Chain of Logic [49]: one prompt that states the rules' semantics and asks for each condition's value and the documents. *Prior-work designs* are Self-consistency, Model executes rules and ReAct. The product ran with three readings, the open rule `any` (Table 2) and a reasoning flag on, which sends a reasoning effort and no temperature. The registered tests ran the full product at the tested commit. Its agents outside the core (Supplement, Section S9) cannot change a scored checklist. The core alone is the registered design *CasePath, checklist only*, which skips those agents; it made 0.34 strict unsafe errors per tenancy claim and 0.55 errors per theft pair, and the tables show both rows. On the frontier model the product ran once, in the checklist-only setting only. That setting skips the document judgments for a claim of one file; every theft text is such a claim, and theft scores requests only. A *case-run* is one run of one design on one case: a *claim-run* on tenancy, a *text-run* on theft; a *pair-run* covers both texts of a theft pair. The other designs ran 3 times per case.

The measures are Study 2's. On tenancy, *strict unsafe errors per claim* count five kinds of error: an *unnecessary request* (a document the reference does not need), a *premature request* (a document

Table 4. Designs of Study 3. Calls and cost per case-run, from the ledgers; tenancy / theft. Cost ratios: against Rules in context on the cost-efficient model (same-model prompt) and on the frontier model (frontier prompt). All designs run on the cost-efficient model unless the name says frontier model; all receive the reviewed rules except the two original-rules designs.

Design	What it is	Calls	USD per case	× same-model prompt	× frontier prompt
CasePath	the product at the frozen commit	28.1 / 30.1	0.0134 / 0.0300	19.8 / 41.8	2.1 / 7.4
CasePath, checklist only	the same commit, run scope `checklist`	8.7 / 4.8	0.0030 / 0.0049	4.4 / 6.8	0.5 / 1.2
Core	two readings and exact deferral outside the product (theft only)	– / 2.0	– / 0.0028	– / 3.9	– / 0.7
Rules in context	one prompt applies the rules	1.0 / 1.0	0.0007 / 0.0007	1.0 / 1.0	0.1 / 0.2
Self-consistency	5 samples of Rules in context; majority per document	5.0 / 5.0	0.0034 / 0.0035	5.0 / 4.9	0.5 / 0.9
Model executes rules	one prompt that states the semantics; the model derives the documents	1.0 / 1.0	0.0007 / 0.0017	1.0 / 2.4	0.1 / 0.4
ReAct	a tool-using agent over the same workflow	8.1 / 2.9	0.0012 / 0.0019	1.8 / 2.6	0.2 / 0.5
Two-valued	CasePath's readings, unresolved taken as false or as true; no calls	–	–	–	–
Rules in context, original rules	the Study 1 rules (theft only)	– / 1.0	– / 0.0008	– / 1.1	– / 0.2
Rules in context, frontier model	the same prompt on the frontier model	1.0 / 1.0	0.0063 / 0.0041	9.3 / 5.7	1.0 / 1.0
original rules	the Study 1 rules on the frontier model (theft only)	– / 1.0	– / 0.0040	– / 5.6	– / 1.0
CasePath, checklist only, frontier model (one run)	the same commit and setting on the frontier model	7.4 / 4.2	0.0214 / 0.0386	31.6 / 53.8	3.4 / 9.5

whose need depends on an open fact), a *missed* document (needed and not requested), a *handed-over* document (needed and passed to a person instead) and a *silent skip* (a document whose need depends on an open fact, marked as not needed). *All deviations per claim* add held documents that should have been requested and *unneeded holds*, documents the reference neither requires nor holds. On theft, *errors per pair* count the document changes between the two texts of a changed-fact pair that are wrong or missing. Unresolved-fact pairs are scored separately: a pair is *unchanged* when the second text's checklist differs from the first only in the open condition's documents, and a *premature request* requests one of the open condition's documents.

6.2 Registered outcomes

Three of five hypotheses were confirmed, one failed and one was not tested (Table 5); Section 6.6 adds a single prompt told the stop and hold rules, which the registered comparators were not. The plan tested them in a fixed order, each at the 0.05 level only if all before it were confirmed: H1 theft and H2 tenancy against Rules in context on the same model; H3 tenancy against Rules in context on the frontier model; H4 theft against each of Self-consistency, Model executes rules and ReAct, all three required; H5 theft non-inferiority to Rules in context on the frontier model, with a margin of 0.25 errors per pair and a paired cluster-bootstrap test. H1 to H4 use exact two-sided sign-flip tests over clusters, the 9 changed conditions on theft and the 24 families on tenancy. Beside H1, the plan's secondary contrast with the original rules does not depend on the comparator's rule base: 0.58 against 1.47 errors per pair ($p = 0.0117$, registered secondary). On the frontier model, the plan's secondary non-inferiority contrast with the original rules (0.58 against 0.67) reports a bootstrap share of 0.0035, which is not a calibrated p; an exact sign-flip test for superiority, added after the test, gives $p = 0.6367$.

H4 failed against Self-consistency. Its exact test gives each of the nine cluster differences a random sign; its smallest attainable p is 0.0039, and it returned 0.0625. CasePath had fewer errors on 7 of

Table 5. Registered hypotheses in the fixed order, with three columns added after the test and not registered. Values per pair (theft) or per claim (tenancy), as means of cluster values. Difference: CasePath minus comparator, with the registered 95% percentile cluster-bootstrap interval (inverted sign-flip intervals: Supplement, Table S24). p: exact two-sided sign-flip test. Added columns: the outcome under the order of the plan's final draft (21:26), under the first draft's order (19:17), and under Holm's correction over the five hypotheses, from the registered p values and, for H5, an exact one-sided sign-flip non-inferiority test.

						Registered	Added after the test, not registered		
Hypothesis and comparator	CasePath	Comp.	Diff.	95% interval	p	Status	Order of 21:26	First draft	Holm (adj. p)
H1 Theft, Rules in context	0.58	1.74	−1.15	−1.65 to −0.64	0.0078	confirmed	confirmed	confirmed	confirmed (0.0234)
H2 Tenancy, Rules in context	0.31	2.14	−1.83	−2.72 to −0.99	0.0002	confirmed	confirmed	not in draft	confirmed (0.0008)
H3 Tenancy, Rules in context, frontier model	0.31	1.86	−1.55	−2.44 to −0.77	0.0008	confirmed	not tested	not tested[a]	confirmed (0.0032)
H4 Theft, Self-consistency	0.58	1.21	−0.63	−1.13 to −0.08	0.0625	not confirmed	not confirmed	not confirmed	not confirmed (0.0625)
Model executes rules	0.58	1.94	−1.36	−1.65 to −1.08	0.0039				
ReAct	0.58	1.69	−1.10	−1.57 to −0.57	0.0117				
H5 Theft, Rules in context, frontier model	0.58	0.95	not tested, because H4 failed			not tested	not tested	not tested	confirmed (0.0234)[b]

[a]The first draft had a non-inferiority form of H3, fourth in its order. [b]H5 was not tested; with its registered bootstrap non-inferiority share, which is not a calibrated p, Holm gives the same statuses (adjusted value for H5 0.0005).

the nine changed conditions and more on 2, by 0.13 and 1.00 errors per pair (expert procedure; third party caused the loss; Supplement, Table S23). 32 of the 512 sign patterns give a sum at least as far from zero. An interval from inverting this test, added after the test, runs from −1.25 to 0.08 and includes zero. With the eight stories as clusters instead, $p = 0.0156$, and H5's superiority p is 0.0547: the unit of clustering matters, and the registered unit stands. H5 was therefore not tested.

The plan reports the tenancy subsets beside H2 and H3 as descriptive family means. As the plan predicted, most of the tenancy difference lies in the 33 denied- and open-event claims: 0.47 strict unsafe errors per claim against 5.69 and 5.31 for the two single prompts. On the 63 ordinary claims, CasePath made 0.20 against 0.44 for the same model ($p = 0.0417$, descriptive; the plan had predicted no difference) and 0.28 for the frontier model ($p = 0.409$). Supplement, Table S17 gives the per-claim values. By family, added after the test, the effect sits in the 9 families that contain denied- or open-event claims (termination and rent-increase disputes), where CasePath was lower in every family by at least 2.33 errors per claim. In the other 15 families (13 defect, 2 termination) the differences are small and mixed: against Rules in context CasePath was lower in 10, equal in 2 and higher in 3, by at most 0.75 (Supplement, Figure S7).

Under Holm's correction over the five hypotheses, which we did not register, H1, H2, H3 and H5 would pass and H4 would not. The registered outcome stands: three confirmed, one failed, one not tested.

6.3 All four cells

Table 6 gives every design in every cell of model and data set, with both tenancy measures. CasePath had fewer errors than Rules in context in three cells and not on theft with the frontier model (Figure 3; registered and registered secondary). On the cost-efficient model it had a lower mean error than every single-prompt and prior-work design in both data sets, and every such

Table 6. All cells of model × data set × design (registered). Tenancy: strict unsafe errors per claim; all deviations per claim, which add held documents that should have been requested and unneeded holds, so that no design gains by holding; share of reference-held documents held. Theft: wrong + missed changes per changed-fact pair; on the 48 unresolved-fact pairs, the share left otherwise unchanged and the share with a premature request. Means over clusters; percent where marked. –: not run.

	Tenancy			Theft		
Design	strict	all dev.	held (%)	errors	unchanged (%)	premature (%)
Cost-efficient model						
CasePath	0.31	0.73	97	0.58	94	0
checklist only (the core)	0.34	0.73	100	0.55	95	0
Core outside the product	–	–	–	0.77	85	0
Rules in context	2.14	2.33	48	1.74	27	1
original rules	–	–	–	1.47	31	1
Self-consistency	2.10	2.31	55	1.21	45	0
Model executes rules	0.83	1.49	86	1.94	27	1
ReAct	1.84	2.00	55	1.69	47	2
Two-valued, unresolved = false	1.27	1.27	0	0.58	94	0
Two-valued, unresolved = true	1.51	1.51	0	2.46	7	65
Frontier model						
CasePath, checklist only (one run)	0.30	0.70	87	0.83	75	0
Rules in context	1.86	2.09	35	0.95	45	8
original rules	–	–	–	0.67	91	2

difference had $p < 0.05$ except the theft difference from Self-consistency (registered tests H1, H2 and H4, and the plan's secondary contrasts S2a–S2c and S4c; Supplement, Table S20). On the frontier model CasePath, checklist only, ran once. Against the frontier prompt it had $p = 0.0012$ on tenancy and $p = 0.672$ on theft (Figure 3). It also made more theft errors than the product on the cost-efficient model with the same rules: 0.83 against 0.58 (+0.25, $p = 0.0781$). These contrasts are registered secondary, descriptive.

An error decomposition, added after the test, attributes each of CasePath's errors to the stage that caused it (Table 7). The run's stored state is replaced stage by stage, in a fixed order, by its designed value; the error that disappears at a stage is that stage's part, and what remains with every stage right is the rule's. The parts add up exactly, and in these data none of them changes with the order of the stages. A part can be negative where correcting one stage uncovers an error another had masked. On tenancy the rule part is zero by construction, because the reference comes from the engine; most of the cost-efficient product's errors are readings that admitted a wrong value, mainly a denied defining event read as having happened. On theft the rule part is 0.333 errors per pair in both cells, between 0.22 and 0.44 under two alternative mappings of designed conditions to rule facts. Supplement, Section S14.1 gives the single prompts' errors by kind.

The full product and the core alone did not differ detectably (registered secondary): CasePath, checklist only, made 0.34 against 0.31 strict unsafe errors per claim ($p = 0.5$) and 0.55 against 0.58 errors per pair ($p = 0.594$), at 22% and 16% of the full product's cost.

6.4 What is held

On tenancy, CasePath held 97% of the documents the reference holds, against 48% and 35% for Rules in context (Table 6; registered). Two-valued execution of its own readings made 1.27 (unresolved taken as false) and 1.51 (taken as true) strict unsafe errors per claim, against 0.31. On theft unresolved-fact pairs, CasePath made no premature request and left 94% of pairs otherwise

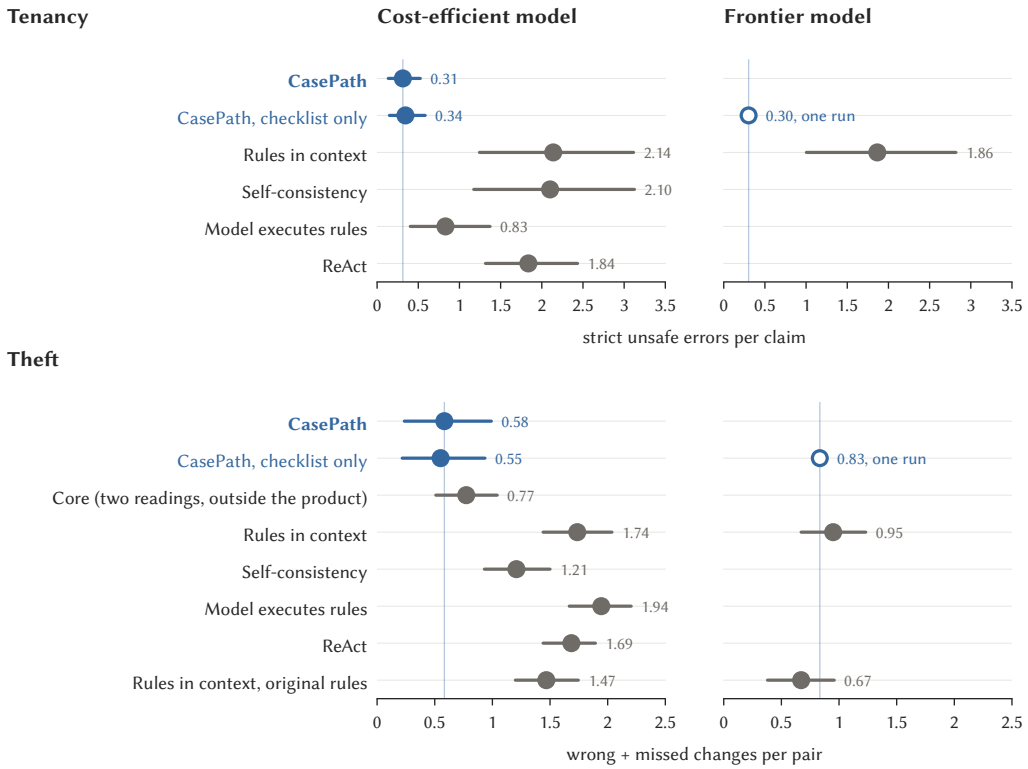


Fig. 3. All four cells of model × data set (registered results). Rows: tenancy, strict unsafe errors per claim; theft, wrong + missed changes per pair; one x scale per row. Columns: the cost-efficient and the frontier model. Filled dot: the design's mean over its clusters (24 families, 9 changed conditions); bar: 95% percentile cluster-bootstrap interval of that mean. Open blue circle: CasePath, checklist only, on the frontier model, one run, without an interval. A row without a mark was not run on that model. Core: two readings and exact deferral run outside the product. Theft on the frontier model: 0.83 against 0.95 (reviewed rules) and 0.67 (original rules).

unchanged, against 27–47% for every single-prompt and prior-work design given the reviewed rules and 91% for the frontier prompt with the original rules (registered). On the share of those pairs in which the open condition's own documents were held, added after the test, CasePath had 83%, Rules in context 85%, the frontier prompt 85% and Self-consistency 90%; the single prompts' errors were changes to other documents.

CasePath also held 0.33 documents per claim that the reference neither requires nor holds, against 0.17 and 0.18 for the single prompts, most of them on denied-event claims where the readings did not all agree on the denial. Counting these as errors, CasePath made 0.73 deviations per claim, against 2.33 ($p = 0.001$) and 2.09 ($p = 0.0023$; registered secondary).

† *Claims with an open fact or a denied or open defining event.* Against Rules in context, the tenancy lead lies in the 56 claims with an open fact (32) or a denied or open defining event (24 denied); the other 40 state every fact (Supplement, Table S39). On the first group, CasePath made 0.40 strict unsafe errors per claim, against 3.39 and 3.06 for Rules in context on the two models. On the claims that state every fact, it made 0.19, against 0.39 for the same-model prompt ($p = 0.2761$) and 0.19 for

Table 7. † CasePath's errors split by the stage that caused them. Means over case-runs (tenancy: strict unsafe errors per claim; theft: errors per pair), which differ from the cluster means of Figure 3 in the third decimal. Cost-efficient: the full product, three runs; frontier: checklist only, one run. Rule: errors with every fact at its designed value (theft: 0.22 to 0.44 under alternative mappings); reading: a wrong value admitted; hold: a stated fact left unresolved or an open fact given a value. –: no such stage on theft.

	Tenancy		Theft	
Stage	cost-efficient	frontier	cost-efficient	frontier
Rule	0.000	0.000	0.333	0.333
Routing	0.045	0.000	–	–
Document judgment	0.076	0.083	–	–
Reading	0.167	0.104	0.088	0.250
Hold	0.024	0.115	0.157	0.250
Other product steps	0.000	0.000	0.005	0.000
Measured (sum)	0.313	0.302	0.583	0.833

the frontier prompt ($p = 0.8984$): no advantage over either. These are means over claim-runs; the p values come from sign-flip tests over family means, which the Supplement, Section S14, gives.

Supplement, Figure S8 draws each design's expected errors against the share of such claims. On the 23 claims with an open branch fact alone, CasePath made 0.23 strict unsafe errors per claim and Rules in context 0.51 (registered subset rows).

6.5 † The same budget

More samples of the same prompt left a large tenancy gap; on theft the gap at matched cost was not significant (Figure 4). We recomputed CasePath, checklist only, with its first $k = 1$, 2 and 3 stored readings, and Self-consistency with $m = 1$ to 15 of its stored samples, without new model calls. On tenancy, Self-consistency stayed near 2.10 strict unsafe errors per claim from $m = 1$ to $m = 15$ (2.10). On theft it fell from 1.84 to about 1.00 errors per pair and levelled off from $m = 9$.

We matched the budget of CasePath, checklist only, not of the full product, because fifteen samples did not reach the full product's cost. The cheapest Self-consistency that cost at least as much as checklist only used $m = 5$ samples on tenancy and $m = 7$ on theft. Checklist only's difference from it was −1.76 strict unsafe errors per claim ($p = 0.0006$) and −0.53 errors per pair ($p = 0.1328$).

More readings lower strict errors and raise unneeded holds. From $k = 1$ to $k = 3$ on tenancy, unneeded holds rose from 0.003 to 0.302 per claim, and all deviations from 0.55 to 0.74.

Cost and time were measured from the ledgers and logs (registered costs). Per tenancy claim and theft text it cost USD 0.013 and 0.030: 19.8 and 41.8 times the same-model single prompt, and 2.1 and 7.4 times the frontier single prompt. Checklist only cost 4.4 and 6.8 times the same-model prompt. On the frontier model, checklist only cost more per tenancy claim (USD 0.0214) than the full product on the cost-efficient model (0.0134). In the test, with 20 cases in parallel, a full case-run took a median of 204 and 287 s, checklist only 33 and 87 s, and the frontier single prompt 14 and 10 s.

6.6 † A prompt that may hold

Told the stop and hold rules in plain words, a single prompt closes the tenancy gap (Table 8; added after the test, on Study 3's data, not registered). The *prompt with hold option* is Rules in context's prompt with its task replaced by a three-way decision per document (request now; hold with a

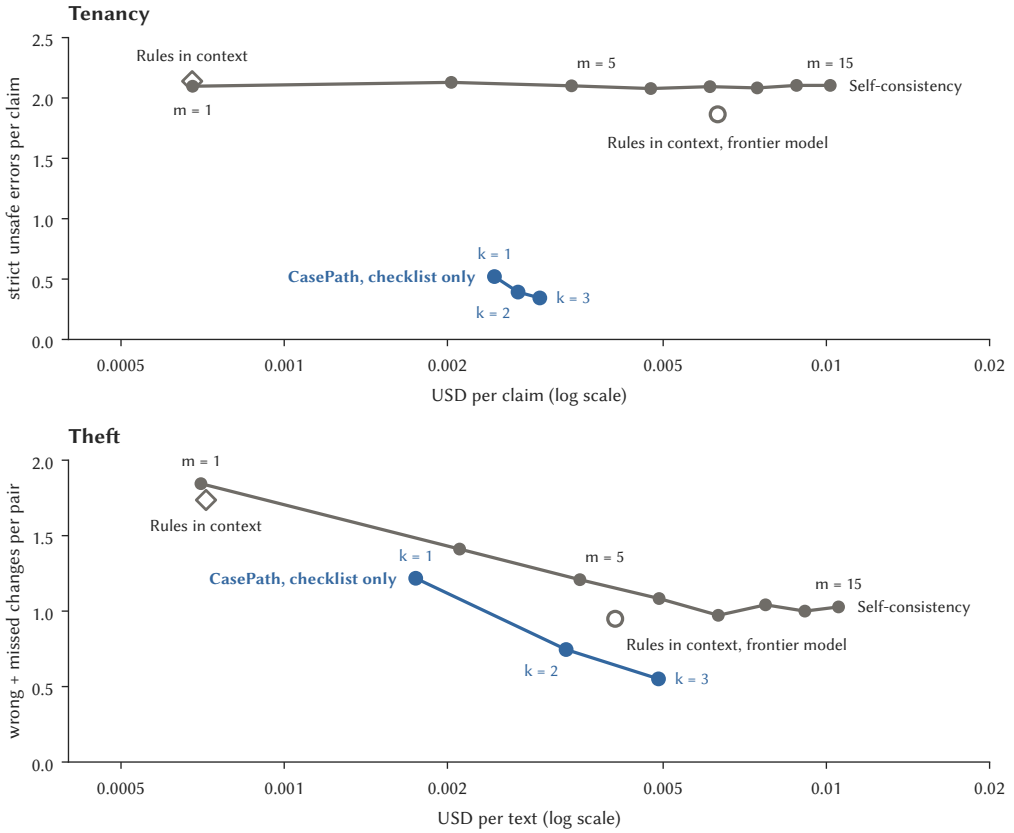


Fig. 4. † Errors against cost per case, log cost axis. Grey line: Self-consistency with $m = 1$ to 15 samples of Rules in context; $m = 1$ is the mean over the 15 stored single samples. Blue: CasePath, checklist only, with $k = 1, 2, 3$ readings. Hollow diamond: Rules in context as registered, whose own runs are different draws from the samples behind $m = 1$. Open circle: Rules in context on the frontier model. Points are means, without intervals. The full product lies to the right of the axis (Table 4).

question when the need depends on a fact the file raises and leaves open; do not request) and with the stop and silence rules the product's workflow encodes. On the frontier model we also ran Model executes rules, whose prompt already had a hold rule. The prompts were hashed before any call; the frontier designs ran once, and none of the 1120 outputs failed. The prompt with hold option made 0.35 strict unsafe errors per claim on the same model as CasePath (0.31; difference −0.03, $p = 0.7379$ over claims). On the frontier model the hold-capable designs made 0.19 and 0.19, against 0.30 for CasePath, checklist only (+0.11, $p = 0.3857$).

On denied defining events CasePath is now the worse design: 0.61 strict unsafe errors and 1.78 deviations per claim, against 0.21 and 1.10 for the prompt on the same model and 0.00 and 0.00 on the frontier model ($p = 0.1250$ and 0.1250 over families). Its readings disagreed on the denial, and it held.

CasePath keeps a lead on all deviations where a branch fact is open and where every fact is stated. On open branch facts it made 0.52 deviations per claim, against 1.25, 1.26 and 1.30 ($p = 0.0127$,

Table 8. † A prompt that may hold (added after the test, on Study 3's data; not registered). Tenancy: strict unsafe errors / all deviations per claim, means over claim-runs, on all 96 claims, the 40 that state every fact, the 23 with an open branch fact only, the 24 with a denied and the 9 with an open defining event. Theft: wrong + missed changes per changed-fact pair. Rules in context rows are Study 3's, for comparison. Supplement, Section S15 gives every contrast.

Design	All	Stated	Open branch	Denied event	Open event	Theft
Cost-efficient model						
CasePath	0.31 / 0.73	0.19 / 0.33	0.23 / 0.52	0.61 / 1.78	0.26 / 0.26	0.58
checklist only	0.34 / 0.73	0.21 / 0.34	0.32 / 0.61	0.72 / 1.78	0.00 / 0.00	0.55
Prompt with hold option	0.35 / 0.89	0.41 / 0.73	0.48 / 1.25	0.21 / 1.10	0.11 / 0.11	1.68
Rules in context (Study 3)	2.14 / 2.33	0.39 / 0.47	0.51 / 0.59	5.68 / 6.24	4.63 / 4.63	1.74
Frontier model						
CasePath, checklist only (one run)	0.30 / 0.70	0.05 / 0.05	0.61 / 0.65	0.29 / 1.79	0.67 / 0.78	0.83
Prompt with hold option (one run)	0.19 / 0.58	0.18 / 0.68	0.48 / 1.26	0.00 / 0.00	0.00 / 0.00	1.18
Model executes rules (one run)	0.19 / 0.54	0.13 / 0.40	0.57 / 1.30	0.00 / 0.25	0.00 / 0.00	1.24
Rules in context (Study 3)	1.86 / 2.09	0.19 / 0.23	0.45 / 0.64	4.43 / 5.06	6.07 / 6.11	0.95

0.0156 and 0.0068 over families). The prompts held documents that were required now when the wording was tentative (0.65 to 0.70 per claim, against 0.19), and they accepted an insufficient original as on file (0.30 to 0.36 missed documents per claim, against 0.16). On files that state every fact, CasePath, checklist only, on the frontier model made 0.05 deviations per claim against 0.68 for the frontier prompt with hold option ($p = 0.0176$ over families).

On theft the hold option does not help the prompt. It made 1.68 errors per pair on the same model, against 0.58 for CasePath ($p = 0.0078$ over the nine clusters), and 1.18 and 1.24 on the frontier model, against 0.83 ($p = 0.0422$ and 0.0196 over pairs; 0.1016 and 0.1484 over clusters). The best frontier single prompt on theft remains Rules in context with the original rules (0.67).

6.7 † Does the result depend on our reading of the rules?

The denied-event claims carry about two thirds of the tenancy lead: 69% of the lead over the same-model prompt and 62% of the lead over the frontier prompt, by claim weight. Amending the workflow to the other reading removes 64% and 36% of the two leads (Table 9).

A clean room checked the engine's implementation. Two AI sessions that never saw the engine, its outputs or the reference were given the rule text, its machine-readable process catalogue and the designed fact states. Each wrote an evaluator, and their references agreed with the registered one on 1024 of 1024 and 1024 of 1024 document decisions. This confirms the implementation of the catalogue.

Then we scored the product's unchanged workflow against references for four readings of the rule text that both evaluators had rejected (A1 to A4). Evaluator B wrote these references afterwards, to our instruction. Under A1, "a denied defining event does not stop the process", the tenancy result reverses: the difference from Rules in context becomes +0.55 ($p = 0.1348$) and from the frontier prompt +0.21 ($p = 0.5134$), and 2 of the 10 registered tenancy contrasts reverse with $p < 0.05$. With all four readings together, 6 of the 10 reverse with $p < 0.05$, among them H2 (+0.94, $p = 0.0234$) and H3 (+0.75, $p = 0.0472$). Under A2, A3 and A4 the result holds with the workflow unchanged; for A2 and A4 these rows are the evidence, because no design was told the reading.

Last, we amended the workflow to each reading and recomputed the checklist from the stored readings, without model calls. The unamended replay reproduces 285 of 286 stored checklists. Under A1, CasePath then keeps a significant but smaller lead. Under A1 the remaining same-model lead rests on the 9 open-event claims.

Table 9. † Sensitivity to the reading of the tenancy rule text. A1: a denied defining event does not stop the process; A2: an open defining event is presumed to have occurred; A3: optional documents are requested; A4: a family-home condition excludes arrears. Agree: document decisions equal to the registered reference (of 1024). Strict unsafe errors per claim, family means: CasePath with the registered workflow (unchanged) and with the workflow amended to the reading (amended); the single prompts' stored checklists scored against the same reference. H2, H3: CasePath minus Rules in context on the cost-efficient and the frontier model, exact sign-flip p in parentheses. Use: the rows that count as evidence; * and ‡: Section 7.1.

		CasePath		Rules in context		Unchanged workflow		Amended workflow		
Reading	Agree	unch.	amend.	cost-eff.	frontier	H2	H3	H2	H3	Use
Registered	1024	0.31	0.31	2.14	1.86	−1.83 (0.0002)	−1.55 (0.0008)	−1.83 (0.0002)	−1.55 (0.0008)	–
A1	865	1.37	0.16	0.82	1.16	0.55 (0.1348)	0.21 (0.5134)	−0.66 (0.0002)	−1.00 (0.0014)	both*
A2	961	0.29	0.29	1.72	1.30	−1.43 (0.0002)	−1.01 (0.0020)	−1.43 (0.0002)	−1.01 (0.0020)	unch.
A3‡	952	1.06	0.34	2.88	2.61	−1.82 (0.0002)	−1.55 (0.0008)	−2.54 (< 0.0001)	−2.27 (< 0.0001)	unch.
A4	1019	0.34	0.31	2.17	1.90	−1.83 (0.0002)	−1.55 (0.0008)	−1.86 (0.0001)	−1.59 (0.0007)	unch.
All four‡	702	2.38	0.14	1.44	1.63	0.94 (0.0234)	0.75 (0.0472)	−1.30 (< 0.0001)	−1.49 (< 0.0001)	unch.

6.8 † Theft on the frontier model

On theft, the decomposition puts the frontier model's extra errors, against the product on the cost-efficient model, in its readings and holds, and the stored readings trace them to two broadly worded conditions. The decomposition splits that cell's 0.833 errors per pair into 0.333 from the rules, 0.250 from readings and 0.250 from holds; the cost-efficient product has the same rule part and 0.088 and 0.157 for the other two (Table 7). One condition requires a third party at fault. The frontier model admitted it as true in 8 of its 8 ordinary runs, on a sentence excluding any third party "apart from the unidentified thief"; the cost-efficient model did so in 0 of its 50 ordinary runs, development runs included. The other concerns traces of the theft that were removed or altered; the frontier model's value for it changed between the two texts of a pair.

We propose a lint as a diagnostic for such conditions; it is a script written after the test and not part of the tested commit. It flags a condition that the readings admit as true in at least half of the runs on ordinary texts, or that at least one reading answers true in three quarters of them. The lint needs no reference answers, but its flags are many. Run on the theft test texts themselves, it flagged both conditions among 14 flags with the frontier model's readings (12 with the cost-efficient model's). On the Study 3 tenancy claims it flags 8 of 10 conditions with the cost-efficient model and 9 of 10 with the frontier model, because those claims were built with several true branch facts each. In a mutation test on tenancy (one model, one run, USD 0.15), the analyst who designed the lint widened 6 conditions to general circumstances; all were flagged, but 3 of them already were in their original wording, and 3 of 5 untouched conditions were flagged too.

Treating the flagged conditions as open did not help out of sample (Supplement, Table S44). With the frontier model's in-sample flags, its theft errors fell from 0.833 to 0.528 per pair, at the price of a larger rule part and more held documents. With flags computed from development runs only, they fell to 0.778, still above the frontier prompt with the original rules; the development flags missed the traces condition. No paired difference is significant (p from 0.1406 to 0.6719 over nine clusters). Two blind AI editor sessions reviewed the flags under the four-eyes rule and could not agree on narrowing the third-party condition: one rejected it, because its source passage states no exclusion.

7 DISCUSSION

Principle 1: state the rules that stop or hold, and keep the holds and document checks in code. On the denied- and open-event tenancy claims, Rules in context made 5.68 and 4.63 strict unsafe errors per claim, Model executes rules, which states the semantics in its prompt, 1.94 and 1.04, and

CasePath 0.61 and 0.26 (registered secondary, descriptive; per-claim means). Added after the test, a prompt told the stop and hold rules made 0.21 and 0.11 (Section 6.6). Stating those rules closed the defining-event gap, in a prompt as well as in code. Code execution's remaining evidence is fewer wrong holds and the document checks: on open branch facts CasePath held 0.19 documents per claim that were required now and missed 0.16, against 0.65 and 0.36 for that prompt.

Principle 2: give an open fact its own value, and evaluate every completion. The two-valued rows support this principle (Section 6.4): unresolved taken as false or as true gave 1.27 and 1.51 strict unsafe errors per claim on CasePath's own readings, against 0.31 (registered). Added after the test: on the 277 of 286 claim-runs where the research engine reproduces CasePath's checklist, exact deferral made 0.25 strict unsafe errors per claim against 0.23 for strong Kleene evaluation of the same facts, and 0.65 against 1.31 deviations. Strong Kleene evaluation had fewer strict errors in 3 claim-runs and more in 0, because it holds more.

Principle 3: admit only agreeing readings with quotations found in the file. Added after the test: few wrong values were admitted (Section 3.5), and with one reading instead of three, strict unsafe errors rose from 0.34 to 0.52 per claim and theft errors from 0.55 to 1.22 per pair (Section 6.5).

For an organisation. Someone must write the workflows and their presumptions, cite a source for each, and have the rules reviewed by a person who can answer for them. An auditor gets each deciding value with its quotation, each hold with its question, and the hash of the code that ran. A hold costs a handler a question to the claimant and a wait for the answer.

7.1 Scope

Appendix B records the run's deviations and irregularities.

- Data: synthetic texts of 77–320 words, written by an AI session that could read the product's prompts and checked by a model of the same family; CasePath alone was developed on earlier data.
- References: tenancy references follow the engine CasePath implements and a silence clause we wrote; the AI clean room confirms the implementation, not the reading; no person checked it; wrong presumptions were only simulated.
- Reading: 69% and 62% of the tenancy lead rest on reading a denied defining event as ending the process, on which no lawyer has ruled; the alternative references follow our instruction; baselines were not re-run (*); rows A3 and "all four" are upper bounds (‡).
- Theft: AI sessions that knew the reused conditions reviewed the rules, without a jurist, and the review worsened both single prompts (1.47 to 1.74, 0.67 to 0.95); CasePath never ran with the original rules; the unchanged share cannot tell a hold from reading the open fact as false.
- Statistics: H3 was tested only after a reorder 38 minutes before the freeze; theft has nine clusters; the frontier product ran once; secondary and post hoc p values are not corrected for multiplicity; two decomposition mappings are uncertain.
- Readings that share one model miss a shared misreading.
- The share of real files with an open fact is unknown.
- Cost: 20–42 times the same-model prompt; times are lower bounds; handler time was not measured.
- The full product and the core shared cached responses.
- The lint was tested only against blunt widening; its false-alarm rate is unknown.
- The prompt with hold option is post hoc, on the data the product was tested on; its frontier designs ran once.

8 CONCLUSION

In a pre-registered test of a frozen commit on synthetic claims, CasePath made 0.31 strict unsafe errors per tenancy claim against 2.14 and 1.86 for single prompts with the same rules, and held 97% of the documents the reference holds against 48% and 35% (registered). A prompt told the same stop and hold rules, added after the test, matched that tenancy result overall; the lead that remains is on open branch facts, insufficient documents and theft. CasePath's theft lead over the same-model prompt was confirmed, and its lead over self-consistency (0.58 against 1.21 errors per pair) gave $p = 0.0625$, so that hypothesis failed (registered). Analyses added after the test show where the lead stops: on files that state every fact; with the product's workflow unchanged, under the reading that a denied defining event does not end the process (amended to that reading, it keeps a smaller lead, -0.66 and -1.00 strict unsafe errors per claim against the two prompts, which rests on the 9 open-event claims); and on theft with the frontier model, where its run once in the checklist-only setting read two broadly worded conditions literally and made 0.83 errors per pair against 0.67 for the frontier prompt with the original rules (registered secondary). A study on an insurer's real files is planned as separate work.

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REFERENCES

[1] Berk Atıl, Sarp Aykent, Alexa Chittams, Lisheng Fu, Rebecca J. Passonneau, Evan Radcliffe, Guru Rajan Rajagopal, Adam Sloan, Tomasz Tudrej, Ferhan Ture, Zhe Wu, Lixinyu Xu, and Breck Baldwin. 2025. Non-Determinism of "Deterministic" LLM System Settings in Hosted Environments. In *Proceedings of the 5th Workshop on Evaluation and Comparison of NLP Systems*, Mousumi Akter, Tahiya Chowdhury, Steffen Eger, Christoph Leiter, Juri Opitz, and Erion Çano (Eds.). Association for Computational Linguistics, Mumbai, India, 135–148. https://doi.org/10.18653/v1/2025.eval4nlp-1.12

[2] Andrew Blair-Stanek and Benjamin Van Durme. 2025. LLMs Provide Unstable Answers to Legal Questions. In *Proceedings of the Twentieth International Conference on Artificial Intelligence and Law (ICAIL 2025)*. Association for Computing Machinery, 425–429. https://doi.org/10.1145/3769126.3769245

[3] Glenn Bruns and Patrice Godefroid. 1999. Model Checking Partial State Spaces with 3-Valued Temporal Logics. In *Computer Aided Verification (11th International Conference, CAV'99, Proceedings) (Lecture Notes in Computer Science)*, Nicolas Halbwachs and Doron Peled (Eds.). Springer, Berlin, Heidelberg, 274–287. https://doi.org/10.1007/3-540-48683-6_25

[4] C. K. Chow. 1970. On Optimum Recognition Error and Reject Tradeoff. *IEEE Transactions on Information Theory* 16, 1 (1970), 41–46. https://doi.org/10.1109/TIT.1970.1054406

[5] Claudio Di Ciccio, Andrea Marrella, and Alessandro Russo. 2015. Knowledge-Intensive Processes: Characteristics, Requirements and Analysis of Contemporary Approaches. *Journal on Data Semantics* 4, 1 (2015), 29–57. https://doi.org/10.1007/s13740-014-0038-4

[6] Marlon Dumas, Fabiana Fournier, Lior Limonad, Andrea Marrella, Marco Montali, Jana-Rebecca Rehse, Rafael Accorsi, Diego Calvanese, Giuseppe De Giacomo, Dirk Fahland, Avigdor Gal, Marcello La Rosa, Hagen Völzer, and Ingo Weber. 2023. AI-augmented Business Process Management Systems: A Research Manifesto. *ACM Transactions on Management Information Systems* 14, 1 (2023), 1–19. https://doi.org/10.1145/3576047

[7] Martin Eling and Martin Lehmann. 2018. The Impact of Digitalization on the Insurance Value Chain and the Insurability of Risks. *The Geneva Papers on Risk and Insurance – Issues and Practice* 43, 3 (2018), 359–396. https://doi.org/10.1057/s41288-017-0073-0

[8] Martin Eling, Davide Nuessle, and Julian Staubli. 2022. The Impact of Artificial Intelligence along the Insurance Value Chain and on the Insurability of Risks. *The Geneva Papers on Risk and Insurance – Issues and Practice* 47, 2 (2022), 205–241. https://doi.org/10.1057/s41288-020-00201-7

[9] Kyle Erwin, Guy Axelrod, Maria Chang, Achille Fokoue, Maxwell Crouse, Soham Dan, Tian Gao, Rosario Uceda-Sosa, Ndivhuwo Makondo, Naweed Khan, and Alexander Gray. 2025. Few-shot Policy (de)composition in Conversational Question Answering. arXiv:2501.11335. https://arxiv.org/abs/2501.11335

[10] European Parliament and Council of the European Union. 2024. Regulation (EU) 2024/1689 of the European Parliament and of the Council of 13 June 2024 Laying Down Harmonised Rules on Artificial Intelligence and Amending Regulations

(EC) No 300/2008, (EU) No 167/2013, (EU) No 168/2013, (EU) 2018/858, (EU) 2018/1139 and (EU) 2019/2144 and Directives 2014/90/EU, (EU) 2016/797 and (EU) 2020/1828 (Artificial Intelligence Act). Official Journal of the European Union, L series, 2024/1689, 12 July 2024. https://eur-lex.europa.eu/eli/reg/2024/1689/oj/eng

[11] Michalis Famelis, Rick Salay, and Marsha Chechik. 2012. Partial Models: Towards Modeling and Reasoning with Uncertainty. In *2012 34th International Conference on Software Engineering (ICSE)*. IEEE, 573–583. https://doi.org/10.1109/ICSE.2012.6227159

[12] Sebastian Farquhar, Jannik Kossen, Lorenz Kuhn, and Yarin Gal. 2024. Detecting Hallucinations in Large Language Models Using Semantic Entropy. *Nature* 630, 8017 (2024), 625–630. https://doi.org/10.1038/s41586-024-07421-0

[13] Luyu Gao, Aman Madaan, Shuyan Zhou, Uri Alon, Pengfei Liu, Yiming Yang, Jamie Callan, and Graham Neubig. 2023. PAL: Program-aided Language Models. In *Proceedings of the 40th International Conference on Machine Learning (Proceedings of Machine Learning Research, Vol. 202)*. PMLR, 10764–10799. https://proceedings.mlr.press/v202/gao23f.html

[14] Tianyu Gao, Howard Yen, Jiatong Yu, and Danqi Chen. 2023. Enabling Large Language Models to Generate Text with Citations. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics, 6465–6488. https://doi.org/10.18653/v1/2023.emnlp-main.398

[15] Yonatan Geifman and Ran El-Yaniv. 2017. Selective Classification for Deep Neural Networks. In *Advances in Neural Information Processing Systems 30 (NIPS 2017)*. https://papers.nips.cc/paper_files/paper/2017/file/4a8423d5e91fda00bb7e46540e2b0cf1-Paper.pdf

[16] Thomas F. Gordon, Henry Prakken, and Douglas Walton. 2007. The Carneades Model of Argument and Burden of Proof. *Artificial Intelligence* 171, 10–15 (2007), 875–896. https://doi.org/10.1016/j.artint.2007.04.010

[17] Shirley Gregor and Alan R. Hevner. 2013. Positioning and Presenting Design Science Research for Maximum Impact. *MIS Quarterly* 37, 2 (2013), 337–355. https://doi.org/10.25300/MISQ/2013/37.2.01

[18] Neel Guha, Julian Nyarko, Daniel Ho, Christopher Ré, Adam Chilton, Aditya K, Alex Chohlas-Wood, Austin Peters, Brandon Waldon, Daniel Rockmore, Diego Zambrano, Dmitry Talisman, Enam Hoque, Faiz Surani, Frank Fagan, Galit Sarfaty, Gregory Dickinson, Haggai Porat, Jason Hegland, Jessica Wu, Joe Nudell, Joel Niklaus, John Nay, Jonathan Choi, Kevin Tobia, Margaret Hagan, Megan Ma, Michael Livermore, Nikon Rasumov-Rahe, Nils Holzenberger, Noam Kolt, Peter Henderson, Sean Rehaag, Sharad Goel, Shang Gao, Spencer Williams, Sunny Gandhi, Tom Zur, Varun Iyer, and Zehua Li. 2023. LegalBench: A Collaboratively Built Benchmark for Measuring Legal Reasoning in Large Language Models. In *Advances in Neural Information Processing Systems 36 (NeurIPS 2023)*. Neural Information Processing Systems Foundation, Inc., 44123–44279. https://doi.org/10.52202/075280-1915

[19] Alan R. Hevner, Salvatore T. March, Jinsoo Park, and Sudha Ram. 2004. Design Science in Information Systems Research. *MIS Quarterly* 28, 1 (2004), 75–106. https://doi.org/10.2307/25148625

[20] Nils Holzenberger and Benjamin Van Durme. 2023. Connecting Symbolic Statutory Reasoning with Legal Information Extraction. In *Proceedings of the Natural Legal Language Processing Workshop 2023*, Daniel Preoţiuc-Pietro, Catalina Goanta, Ilias Chalkidis, Leslie Barrett, Gerasimos Spanakis, and Nikolaos Aletras (Eds.). Association for Computational Linguistics, Singapore, 113–131. https://doi.org/10.18653/v1/2023.nllp-1.12

[21] Ronald Howard. 1966. Information Value Theory. *IEEE Transactions on Systems Science and Cybernetics* 2, 1 (1966), 22–26. https://doi.org/10.1109/TSSC.1966.300074

[22] Richard Hull. 2008. Artifact-Centric Business Process Models: Brief Survey of Research Results and Challenges. In *On the Move to Meaningful Internet Systems: OTM 2008 (OTM 2008 Confederated International Conferences, Proceedings, Part II) (Lecture Notes in Computer Science)*, Robert Meersman and Zahir Tari (Eds.). Springer, Berlin, Heidelberg, 1152–1163. https://doi.org/10.1007/978-3-540-88873-4_17

[23] Tomasz Imieliński and Witold Lipski, Jr. 1984. Incomplete Information in Relational Databases. *J. ACM* 31, 4 (1984), 761–791. https://doi.org/10.1145/1634.1886

[24] William Jurayj, Nils Holzenberger, and Benjamin Van Durme. 2026. Language Models and Logic Programs for Trustworthy Tax Reasoning. *Proceedings of the AAAI Conference on Artificial Intelligence* 40, 45 (2026), 38688–38698. https://doi.org/10.1609/aaai.v40i45.41212

[25] Amita Kamath, Robin Jia, and Percy Liang. 2020. Selective Question Answering under Domain Shift. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, Dan Jurafsky, Joyce Chai, Natalie Schluter, and Joel Tetreault (Eds.). Association for Computational Linguistics, Online, 5684–5696. https://doi.org/10.18653/v1/2020.acl-main.503

[26] Timotheus Kampik, Christian Warmuth, Adrian Rebmann, Ron Agam, Lukas N. P. Egger, Andreas Gerber, Johannes Hoffart, Jonas Kolk, Philipp Herzig, Gero Decker, Han van der Aa, Artem Polyvyanyy, Stefanie Rinderle-Ma, Ingo Weber, and Matthias Weidlich. 2025. Large Process Models: A Vision for Business Process Management in the Age of Generative AI. *KI – Künstliche Intelligenz* 39, 2 (2025), 81–95. https://doi.org/10.1007/s13218-024-00863-8

[27] Manuj Kant, Manav Kant, Marzieh Nabi, Preston Carlson, and Megan Ma. 2024. Equitable Access to Justice: Logical LLMs Show Promise. arXiv preprint arXiv:2410.09904. arXiv:2410.09904 [cs.AI] https://arxiv.org/abs/2410.09904

[28] Myung Ho Kim. 2026. Evidence-Obligation Pool-Gated Retrieval: Stable Multi-Cycle Retrieval via Evidence Ledger and Warrant Judge. Preprints.org, version 1, posted 15 July 2026. https://doi.org/10.20944/preprints202607.1060.v1 Preprint (not peer reviewed).

[29] Stephen Cole Kleene. 1952. *Introduction to Metamathematics.* D. Van Nostrand Company.

[30] Fabian König, Andreas Egger, Wolfgang Kratsch, Maximilian Röglinger, and Niklas Wördehoff. 2025. Unstructured Data in Process Mining: A Systematic Literature Review. *ACM Transactions on Management Information Systems* 16, 3 (2025), 1–34. https://doi.org/10.1145/3727148

[31] Lorenz Kuhn, Yarin Gal, and Sebastian Farquhar. 2023. Semantic Uncertainty: Linguistic Invariances for Uncertainty Estimation in Natural Language Generation. In *The Eleventh International Conference on Learning Representations (ICLR 2023).* https://openreview.net/forum?id=VD-AYtP0dve

[32] Navish Kumar, Tobias Hoesli, Aydin Javadov, and Joseph Ollier. 2026. CasePath: Source Code, Datasets, Experiment Harness and Analysis Plan. GitHub repository. https://github.com/KumarNavish/casepath Experiments and paper source in `research/tmis2026`.

[33] Dongchen Li, Zhuo Jin, Linyi Qian, and Hailiang Yang. 2025. Textual Analysis of Insurance Claims with Large Language Models. *Journal of Risk and Insurance* 92, 2 (2025), 505–535. https://doi.org/10.1111/jori.70004

[34] Shuyue Stella Li, Vidhisha Balachandran, Shangbin Feng, Jonathan S. Ilgen, Emma Pierson, Pang Wei Koh, and Yulia Tsvetkov. 2024. MediQ: Question-Asking LLMs and a Benchmark for Reliable Interactive Clinical Reasoning. In *Advances in Neural Information Processing Systems 37 (NeurIPS 2024).* Neural Information Processing Systems Foundation, Inc., 28858–28888. https://doi.org/10.52202/079017-0908

[35] Zekun Li, Shinda Huang, Jiangtian Wang, Nathan Zhang, Antonis Antoniades, Wenyue Hua, Kaijie Zhu, Sirui Zeng, Chi Wang, William Yang Wang, and Xifeng Yan. 2026. SOPBench: Evaluating Language Agents at Following Standard Operating Procedures and Constraints. In *Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing*, Hung-Yi Lee, Sunipa Dev, Junyi Jessy Li, and Desmond Elliott (Eds.). Association for Computational Linguistics, Budapest, Hungary, 21561–21616. https://doi.org/10.18653/v1/2026.emnlp-main.936

[36] Denis Merigoux, Nicolas Chataing, and Jonathan Protzenko. 2021. Catala: A Programming Language for the Law. *Proceedings of the ACM on Programming Languages* 5, ICFP (2021), 1–29. https://doi.org/10.1145/3473582

[37] James Mohun and Alex Roberts. 2020. *Cracking the Code: Rulemaking for Humans and Machines.* OECD Working Papers on Public Governance 42. OECD Publishing. https://doi.org/10.1787/3afe6ba5-en

[38] Object Management Group. 2016. Case Management Model and Notation (CMMN), Version 1.1. OMG formal specification, OMG Document Number formal/2016-12-01. https://www.omg.org/spec/CMMN/1.1

[39] Object Management Group. 2024. Decision Model and Notation (DMN), Version 1.5. OMG formal specification, OMG Document Number formal/24-01-01. https://www.omg.org/spec/DMN/1.5

[40] Theo Olausson, Alex Gu, Ben Lipkin, Cedegao Zhang, Armando Solar-Lezama, Joshua Tenenbaum, and Roger Levy. 2023. LINC: A Neurosymbolic Approach for Logical Reasoning by Combining Language Models with First-Order Logic Provers. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, Houda Bouamor, Juan Pino, and Kalika Bali (Eds.). Association for Computational Linguistics, Singapore, 5153–5176. https://doi.org/10.18653/v1/2023.emnlp-main.313

[41] Liangming Pan, Alon Albalak, Xinyi Wang, and William Wang. 2023. Logic-LM: Empowering Large Language Models with Symbolic Solvers for Faithful Logical Reasoning. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, Houda Bouamor, Juan Pino, and Kalika Bali (Eds.). Association for Computational Linguistics, Singapore, 3806–3824. https://doi.org/10.18653/v1/2023.findings-emnlp.248

[42] Ken Peffers, Tuure Tuunanen, Marcus A. Rothenberger, and Samir Chatterjee. 2007. A Design Science Research Methodology for Information Systems Research. *Journal of Management Information Systems* 24, 3 (2007), 45–77. https://doi.org/10.2753/MIS0742-1222240302

[43] M. Pesic and W. M. P. van der Aalst. 2006. A Declarative Approach for Flexible Business Processes Management. In *Business Process Management Workshops (BPM 2006 International Workshops, Proceedings) (Lecture Notes in Computer Science)*, Johann Eder and Schahram Dustdar (Eds.). Springer, Berlin, Heidelberg, 169–180. https://doi.org/10.1007/11837862_18

[44] Raymond Reiter. 1978. On Closed World Data Bases. In *Logic and Data Bases*, Hervé Gallaire and Jack Minker (Eds.). 55–76. https://doi.org/10.1007/978-1-4684-3384-5_3

[45] R. Reiter. 1980. A Logic for Default Reasoning. *Artificial Intelligence* 13, 1–2 (1980), 81–132. https://doi.org/10.1016/0004-3702(80)90014-4

[46] Marzieh Saeidi, Max Bartolo, Patrick Lewis, Sameer Singh, Tim Rocktäschel, Mike Sheldon, Guillaume Bouchard, and Sebastian Riedel. 2018. Interpretation of Natural Language Rules in Conversational Machine Reading. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing.* Association for Computational Linguistics, Brussels, Belgium, 2087–2097. https://doi.org/10.18653/v1/D18-1233

[47] Maung K. Sein, Ola Henfridsson, Sandeep Purao, Matti Rossi, and Rikard Lindgren. 2011. Action Design Research. *MIS Quarterly* 35, 1 (2011), 37–56. https://doi.org/10.2307/23043488

[48] M. J. Sergot, F. Sadri, R. A. Kowalski, F. Kriwaczek, P. Hammond, and H. T. Cory. 1986. The British Nationality Act as a Logic Program. *Commun. ACM* 29, 5 (1986), 370–386. https://doi.org/10.1145/5689.5920

[49] Sergio Servantez, Joe Barrow, Kristian Hammond, and Rajiv Jain. 2024. Chain of Logic: Rule-Based Reasoning with Large Language Models. In *Findings of the Association for Computational Linguistics: ACL 2024*, Lun-Wei Ku, Andre Martins, and Vivek Srikumar (Eds.). Association for Computational Linguistics, Bangkok, Thailand, 2721–2733. https://doi.org/10.18653/v1/2024.findings-acl.159

[50] Bas C. van Fraassen. 1966. Singular Terms, Truth-Value Gaps, and Free Logic. *The Journal of Philosophy* 63, 17 (1966), 481–495. https://doi.org/10.2307/2024549

[51] Maxim Vidgof, Stefan Bachhofner, and Jan Mendling. 2023. Large Language Models for Business Process Management: Opportunities and Challenges. In *Business Process Management Forum (BPM 2023) (Lecture Notes in Business Information Processing)*, Chiara Di Francescomarino, Andrea Burattin, Christian Janiesch, and Shazia Sadiq (Eds.). Springer, 107–123. https://doi.org/10.1007/978-3-031-41623-1_7

[52] Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc V. Le, Ed H. Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. 2023. Self-Consistency Improves Chain of Thought Reasoning in Language Models. In *International Conference on Learning Representations (ICLR 2023)*. https://openreview.net/forum?id=1PL1NIMMrw

[53] Bingbing Wen, Jihan Yao, Shangbin Feng, Chenjun Xu, Yulia Tsvetkov, Bill Howe, and Lucy Lu Wang. 2025. Know Your Limits: A Survey of Abstention in Large Language Models. *Transactions of the Association for Computational Linguistics* 13 (2025), 529–556. https://doi.org/10.1162/tacl_a_00754

[54] Ruixuan Xiao, Wentao Ma, Ke Wang, Yuchuan Wu, Junbo Zhao, Haobo Wang, Fei Huang, and Yongbin Li. 2024. FlowBench: Revisiting and Benchmarking Workflow-Guided Planning for LLM-based Agents. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, Yaser Al-Onaizan, Mohit Bansal, and Yun-Nung Chen (Eds.). Association for Computational Linguistics, Miami, Florida, USA, 10883–10900. https://doi.org/10.18653/v1/2024.findings-emnlp.638

[55] Shunyu Yao, Noah Shinn, Pedram Razavi, and Karthik Narasimhan. 2025. τ-bench: A Benchmark for Tool-Agent-User Interaction in Real-World Domains. In *International Conference on Learning Representations (ICLR 2025)*. https://openreview.net/forum?id=roNSXZpUDN

[56] Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. 2023. ReAct: Synergizing Reasoning and Acting in Language Models. In *International Conference on Learning Representations (ICLR 2023)*. https://openreview.net/forum?id=WE_vluYUL-X

[57] Xi Ye, Qiaochu Chen, Isil Dillig, and Greg Durrett. 2023. SatLM: Satisfiability-Aided Language Models Using Declarative Prompting. In *Advances in Neural Information Processing Systems 36 (NeurIPS 2023)*. Neural Information Processing Systems Foundation, Inc., 45548–45580. https://doi.org/10.52202/075280-1974

[58] Xuan Zhang, Yang Deng, Zifeng Ren, See-Kiong Ng, and Tat-Seng Chua. 2024. Ask-before-Plan: Proactive Language Agents for Real-World Planning. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, Yaser Al-Onaizan, Mohit Bansal, and Yun-Nung Chen (Eds.). Association for Computational Linguistics, Miami, Florida, USA, 10836–10863. https://doi.org/10.18653/v1/2024.findings-emnlp.636

A PROOFS

Proposition 3.1. Strong Kleene connectives are monotone in the information order in which U lies below both T and F [29]. Equations (1) and (2) compose such connectives over the acyclic graph, so by induction in topological order every value in {T, F} persists in every completion. The proof needs the two scope conditions of the statement. The equations must be the workflow's semantics, which holds for flow workflows: outside them, a relation that suppresses or blocks a step can turn an active step inactive when a fact becomes known, and the proposition can fail, for instance when a suppressing transition under an open fact enters a step that is also blocked. Completions must range over {T, F} for every fact independently.

The tested workflows are flow workflows. We checked this for the four workflows of Study 3 with the product's own code at commit 6a72159c (`process/workflows.py`). The product walks one path: at each step it follows the first transition whose condition holds. Equation (1) activates every step reached through a transition that holds. For each tenancy workflow we enumerated every complete assignment of the facts its conditions use (32, 32 and 8) and compared the documents the product's

walk lists with those of Equations (1) and (2); all agreed. In one step of the defect workflow two transitions can hold together, and both lead to the same later step, so the two semantics activate the same steps. For every three-valued assignment (243, 243 and 27) the product's evaluation also requested and held exactly what Equation (3) gives. Optional documents were left out of both comparisons: the product lists them but never requests or holds them. The theft workflow is one step without transitions, so Equation (1) activates only that step and Equation (2) reduces to each document's own condition. `analysis/check_fragment.py` repeats these comparisons. Before the run, the driver's preflight check compared the product's evaluator with the reference engines (Supplement, Section S11).

The plain walk on theft. Above eight open facts, or 256 combinations of answers, the product returns the plain walk, which evaluates each condition in strong Kleene logic and stops at the first transition whose condition is open. Every theft case-run of Study 3 was evaluated this way (384 of 384 for CasePath). On the theft workflow this changes nothing. It has no transitions, and each of its document conditions is empty, one fact tested for true, or a disjunction of distinct facts each tested for true. Strong Kleene evaluation of such a formula is T if a disjunct is T, F if all are F, and U otherwise; U is exactly the case in which two completions disagree. On 5,000 random three-valued assignments the product's evaluation and Equation (3) agreed on every document (`analysis/check_fragment.py`). This check differs from the preflight check described in the Supplement, Section S11, which compared the product's evaluator with the reference engine on 3,000 random theft assignments.

Proposition 3.2. In the tenancy defect workflow, a flow workflow over two-valued facts, the intake step branches to a safety step when health effects are alleged and directly to the timeline step otherwise; the safety step leads to the timeline step as well. With $v(\mathit{health}) = \mathsf{U}$, Equation (1) gives $a(\mathit{timeline}) = (\mathsf{U} \wedge \mathsf{T}) \vee (\mathsf{T} \wedge \mathsf{U}) = \mathsf{U}$, so every document of the timeline step and of the later steps it reaches is held, unless its own or its branch condition is F. Yet under both completions $a(\mathit{timeline}) = \mathsf{T}$.

Proposition 3.3. The observation before it follows from Equation (3): $\hat{r}(d) = \mathsf{U}$ exactly when two completions disagree on $r(d)$, and a value in $\{\mathsf{T}, \mathsf{F}\}$ is the value under every completion. For the proposition: if $r(d) \neq \mathsf{U}$, Proposition 3.1, which needs a flow workflow, gives the same value under every completion, so $\hat{r}(d) = r(d)$.

Proposition 3.4. (a) Under the assumption, v^* is a completion of v. If $\hat{r}_v(d) = \mathsf{T}$, every completion needs d, so $r_{v^*}(d) = \mathsf{T}$ and the request is needed; if $\hat{r}_v(d) = \mathsf{F}$, then $r_{v^*}(d) = \mathsf{F}$ and marking d not needed is right. The remaining documents are held, which is not an unsafe error. Whether a needed document is on file is taken from the document judgments; a wrong judgment there is outside the proposition. (b) The checklist is computed from v alone, so a wrong value enters the evaluation exactly as a correct one would. A document with $\hat{r}_v(d) = \mathsf{T}$ and $r_{v^*}(d) = \mathsf{F}$ is an unnecessary request; one with $\hat{r}_v(d) = \mathsf{F}$ and $r_{v^*}(d) = \mathsf{T}$ is missed.

Exact computation of reading errors. To quantify both kinds without depending on any model, we computed exactly, for all 150 tenancy claims of Study 1 and every partial assignment of their facts, which documents each semantics requests and holds (`scicore/s1_propagation.py`). The noise model judges each fact of a claim correctly with probability $1 - \varepsilon$ and misjudges it otherwise, independently of the other facts; a misjudgement is unresolved with probability ρ and the opposite value with probability $1 - \rho$. We report $\varepsilon = 0.1$ with $\rho = 1$ (every misjudgement unresolved) and $\rho = 0$ (every misjudgement a wrong value). The expectations are exact sums over all assignments, without sampling. Counts are expected numbers of documents summed over the claims; unnecessary

requests are counted beyond those made with no misjudgement. Of the 1002 documents that the claims need, unresolved values become 84.5 expected holds under exact deferral and no unsafe error. Strong Kleene evaluation holds 68% more; reading unresolved values as false misses 84.5 documents, and reading them as true adds 39.5 unnecessary requests. Wrong values cause 84.5 missed documents and 37.0 unnecessary requests under every semantics. The value 84.5 appears three times for one reason: for every pattern of misjudged facts in these claims, the required documents exact deferral holds when the facts are unresolved are exactly those lost when the facts are read as false, and exactly those lost when they take the opposite value. Each pattern has the same probability in the three computations, so the three expectations are equal. Figure 5 shows the same quantities per 100 required documents.

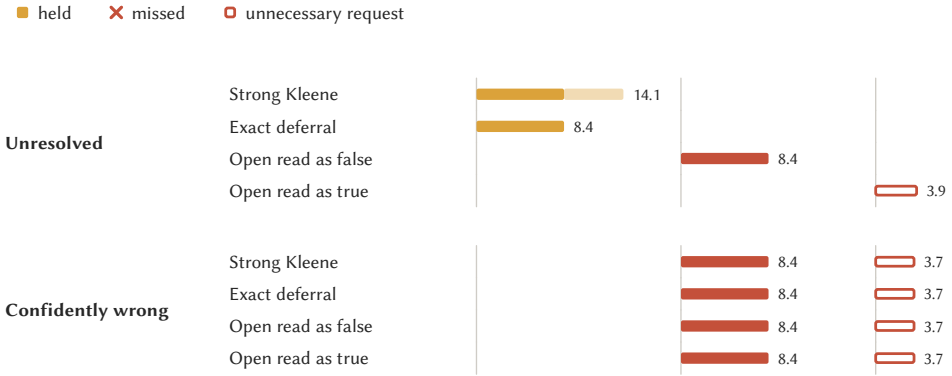


Fig. 5. What reading errors become, computed exactly over all 150 tenancy claims of Study 1, with each fact misjudged with probability 0.1: above, every misjudgement unresolved; below, every misjudgement the opposite value. Bars give expected documents per 100 of the 1002 required documents, that is, percent. Under strong Kleene evaluation the light part of the bar is the holds beyond those of exact deferral. Unnecessary requests are counted beyond those made with no misjudgement. Three-valued evaluation turns unresolved values into holds, two-valued evaluation turns them into missed or unnecessary requests, and a wrong value becomes an error under every semantics.

B DEVIATIONS AND IRREGULARITIES OF STUDY 3

Each item below is stated in one line; the Supplement, Section S11, gives every item in full. The plan (`ANALYSIS_PLAN_STUDY3.md`) was committed before any design ran on `fresh2/`, and the analysis was run once on the stored outputs. Studies 1 and 2 are covered in the Supplement, Section S3.

The run.

- The run needed 5 invocations of the driver; 4 stopped early, each resumed from the response cache with product, model, data and analysis unchanged.
- The first was interrupted by the operator to add parallel workers; its log line ("interrupted during the checks", with a wrong spend) is wrong and is corrected by a dated note.
- The second was killed by the session's time limit on background commands; the wrapper's interrupt had no effect.
- The third ended when the container was restarted; disk, cache and worktree were unchanged.
- The fourth was stopped, as planned, by the wrapper before the time limit; the fifth completed the run.

- Workers rose from 6 to 20 after the first invocation, and the main spending cap was lowered at each resumption to what remained of USD 32; the plan says settings stay unchanged.
- The ledgers record USD 34.33 for the run (22.41 cost-efficient, 11.92 frontier model).
- 50 comparator outputs failed: 43 design outputs, scored as empty checklists as planned (40 from ReAct, none from a product design), and 7 Self-consistency samples, counted as empty votes; a post-hoc analysis leaves ReAct's failures out.
- No model snapshot, temperature or seed is recorded; the provider's defaults applied.

Before the run.

- The run started before the product's owners confirmed the merge of the tested commit; at the time of writing they have not confirmed it.
- The test texts were written by an AI agent session with read access to the product's prompts and development results (Appendix C).
- The probe claims were written knowing the test set's design but not its texts; the plan's "writer that had not seen `fresh2/`" holds for the texts only.
- The hypotheses were reordered 38 minutes before the freeze, on development evidence (Section 6.2).
- The open rule overruled a criterion on wrong holds that the directing author had set earlier.
- Two comparator texts changed during Study 2's development; Model executes rules and ReAct were written during the product's development and run on development data before the freeze.

Departures from the plan's text.

- Calls per case differ from the plan's design table (CasePath about 26 planned, 28.1 and 30.1 measured).
- Self-consistency costs less than CasePath, checklist only, on theft, so no design is matched in cost as the plan assumed.
- The full product and checklist only share cached responses, so their contrast is not between independent runs.
- The Core's runs share readings and are not independent.
- Cost per case joins each call to the ledger by its cache key; some agent costs could not be attributed.
- The subsets were selected from the reference; their counts equal the plan's.
- Subset rows beside H2 and H3 are family-mean contrasts, marked descriptive.
- 2 secondary contrasts are not listed in the plan.
- H5's non-inferiority p is reported outside the procedure.
- The plan fixes the number of resamples; the specification fixes the seed.
- Two-valued with unresolved taken as false equals CasePath on theft, because theft scores requests only.

After the results.

- Thirteen analyses were added after a first referee report and claims audit, and further checks after an outside review; none is registered or changes a status (Supplement, Sections S13.2 and S14).
- From the end of the test, the theft stories of `fresh2/` are development data; the diagnosis and the lint used them.
- The lint's mutation test made the only new model calls (USD 0.15).

- One clean-room evaluator listed a scratch directory after delivering its hashed reference and saw other file names; by its report it opened none.
- Changes for speed live on a separate branch, not the tested commit.

C REPRODUCIBILITY AND USE OF GENERATIVE AI

Release. The repository [32] will hold, once the release choice below is made, the tested product at commit 6a72159c, whose parity command prints the source-tree hash `03048020. . .`; the research workflows; the synthetic data of all three studies; the harness, the analysis scripts and every stored checklist, ledger and run log; and the frozen plans. The Study 3 response cache (about 23,000 files) will be offered as a separate archive. At the tested commit the source names the insurer's subsidiary as the default organisation, so the public release is either a scrubbed commit with its own parity hash or the tested commit with the insurer's approval. The insurer's workflows and Knowledge are not released. Supplement, Section S17 gives the details and the licences.

Reproducing without model calls. One command, `reproduce.sh`, re-checks the stored results from a clone without model calls: data manifests, the parity hash, the claims of Appendix A, Study 3's registered analysis, the analyses added after review and every number macro, byte for byte. Several analyses added after review re-read the product's response cache (the readings and cost split of `study3_review.py`; `study3_feedback.py`, `decomposition.py` and `lint_followup.py`). With the cache archive in place they are re-run and compared byte for byte; without it, the command regenerates their macros from the stored result files, runs the tests that need no cache, and reports the re-runs as skipped. It does not re-check the model runs, whose stored outputs are its inputs. A re-run with model calls needs the reader's own model endpoint, and its answers may differ.

Generative AI. The two evaluated models are the objects of study. An AI agent session wrote the test texts, and sessions of another model of the same family checked them blind. AI editor sessions reviewed the theft rules and the lint's flags; AI sessions wrote the clean-room references. AI coding and writing assistants helped implement the harness and the product changes, run the analyses and draft the text, under the authors' direction; the authors are responsible for all of it.